# Mechanical energy for a gradient-corrected restart system

**Abstract**

We study a two-step gradient iteration with a gradient-difference correction and a varying extrapolation coefficient. For a strongly convex smooth potential, the usual potential-plus-kinetic energy decreases for every coefficient sequence in a fixed interval. We prove a uniform two-step contraction, including the boundary step at which the one-step gradient dissipation vanishes. Adding two consecutive energy values gives an explicit coefficient-indexed family which contracts across every permitted switch. Consequently, both physical and extrapolated trajectories converge geometrically and the physical increments have finite total length. As a corollary, we verify these conclusions for the extended-speed clock-reset rule of Maulén, Guo and Peypouquet, under its experimental gradient-correction normalization and a documented initialization convention. The energy and the finite-step summation are classical; the purpose is to verify their consequences for the stated map, without a dwell-time assumption or a physical-state reset.

**Keywords:** Lyapunov function; inertial iteration; Hessian-driven damping; finite-step decrease; trajectory convergence

**2020 Mathematics Subject Classification.** Primary 39A30; Secondary 37B25, 65K10.

# 1 Introduction

An inertial iteration may change its extrapolation coefficient in response to the trajectory. A Lyapunov function which depends only on the physical phase is useful in this setting: its decrease can be checked before determining whether a switch occurs. A one-step decrease need not be strict, however, even for a strongly convex potential. The boundary gradient step provides a simple instance of this distinction.

Maulén, Guo and Peypouquet [1] introduce an extended-speed restart for a continuous inertial system with Hessian-driven damping. Their Section 7.2 gives a discrete clock-reset heuristic, while Section 8 leaves its convergence analysis untreated. In that discrete rule the new point is retained and only the clock is reset. Its history therefore differs from the state-reset or rejected-trial conventions of other restart methods. The present note treats the normalization used in their experiments and derives the clock-rule conclusion from a more general coefficient-independent energy estimate.

The ingredients are elementary. We use the smooth descent inequality, strong convexity and a segment-averaged Hessian bound. Closely related descent inequalities and energy arguments occur in Attouch, Chbani, Fadili and Riahi [2], and in the inertial analyses of Wang and Peypouquet [3, 4]. The mechanical energy is classical [8], and summing consecutive values is a standard way to turn a finite-step decrease into a one-step decrease. Our result is a verification for the exact recurrence below. It does not propose

a new restart algorithm or establish an advantage of restarting over the same recurrence without restarts.

Let $f : \mathbb{R}^d \to \mathbb{R}$ be twice continuously differentiable, where $d \geq 1$ is finite, and suppose that

$$0 < \mu \leq L, \qquad \mu I \preceq \nabla^2 f(x) \preceq LI \quad \text{for every } x \in \mathbb{R}^d. \tag{1}$$

Write $z$ for its unique minimizer and $f_* = f(z)$. Fix $s > 0$ and $\theta_0 \in (0, 1]$ such that $sL \leq 1$. Starting from arbitrary $x_0, x_1 \in \mathbb{R}^d$, choose any sequence

$$\theta_k \in [\theta_0, 1], \qquad k \geq 1, \tag{2}$$

and define, at the original indices,

$$\begin{aligned} d_k &= x_k - x_{k-1}, \\ y_k &= x_k + \theta_k d_k - s\big(\nabla f(x_k) - \nabla f(x_{k-1})\big), \\ x_{k+1} &= y_k - s\nabla f(y_k). \end{aligned} \tag{3}$$

The coefficients may depend discontinuously on the past trajectory. No condition relating consecutive coefficients is imposed.

Put

$$\begin{gathered} \gamma = s\mu, \qquad r = \max\{1 - \gamma, 1 - \theta_0\}, \\ a = \frac{1 - (1 - \gamma)r^2}{2}, \qquad b = \frac{s^2(1 - sL)}{2}, \\ C = \max\left\{\frac{1}{2} + \frac{4r^2}{\gamma}, \frac{1}{\gamma}\right\}, \qquad \rho = \frac{a}{C}, \qquad q = 1 - \rho. \end{gathered} \tag{4}$$

Thus $0 < \gamma \leq 1$, $0 \leq r < 1$, $a > 0$, $b \geq 0$ and $0 < \rho \leq 1/2$, so $1/2 \leq q < 1$. For an independent phase pair $(x, p)$, define

$$E(x, p) = s\big(f(x) - f_*\big) + \frac{1}{2}\|x - p\|^2. \tag{5}$$

Let $T_\theta(x, p)$ be the new physical point obtained by one step of (3), with $x_k = x$, $x_{k-1} = p$ and coefficient $\theta$. Define

$$\mathcal{V}_\theta(x, p) = E(x, p) + E\big(T_\theta(x, p), x\big). \tag{6}$$

This is a finite evaluation at the current phase and coefficient. It contains no future tail. The energy $E$ is a single common function; $\mathcal{V}_\theta$ is a coefficient-indexed family.

**Theorem 1.** *Under (1)–(3), all iterates exist. For every independent phase pair,*

$$\frac{\gamma}{6}\big(\|x - z\|^2 + \|p - z\|^2\big) \leq E(x, p) \leq \frac{3}{2}\big(\|x - z\|^2 + \|p - z\|^2\big). \tag{7}$$

*Writing $E_k = E(x_k, x_{k-1})$, we have, for every $k \geq 1$,*

$$E_{k+1} - E_k \leq -a\|d_k\|^2 - b\|\nabla f(y_k)\|^2, \tag{8}$$

$$E_k \leq C\big(\|d_k\|^2 + \|d_{k+1}\|^2\big), \tag{9}$$

$$E_{k+2} \leq qE_k. \tag{10}$$

*Moreover, for every $\theta, \eta \in [\theta_0, 1]$,*

$$\mathcal{V}_\eta\big(T_\theta(x,p), x\big) \le (1-\rho/2)\mathcal{V}_\theta(x,p), \tag{11}$$

*and, uniformly in $\theta$,*

$$\frac{\gamma}{6}\big(\|x-z\|^2 + \|p-z\|^2\big) \le \mathcal{V}_\theta(x,p) \le 3\big(\|x-z\|^2 + \|p-z\|^2\big). \tag{12}$$

*The original trajectory satisfies*

$$E_k \le q^{\lfloor (k-1)/2 \rfloor} E_1 \qquad (k \ge 1), \tag{13}$$

*with*

$$\begin{aligned} \|x_k - z\|^2 \le \frac{2E_k}{\gamma}, \qquad \|d_k\|^2 \le 2E_k, \\ \|y_k - z\|^2 \le 4(1/\gamma + r^2)E_k, \qquad f(x_k) - f_* \le \frac{E_k}{s}. \end{aligned} \tag{14}$$

*In particular, $x_k$ and $y_k$ converge geometrically to $z$, and*

$$\sum_{k=1}^{\infty} \|x_k - x_{k-1}\| \le \frac{2\sqrt{2E_1}}{1-\sqrt{q}}. \tag{15}$$

*All conclusions include $sL = 1$ and zero initial energy.*

The estimates are uniform over the coefficient sequence. Their constants may depend on $s, \mu, L$ and $\theta_0$; no optimal or accelerated condition-number rate is asserted. The theorem does not require a monotone coefficient sequence, a minimum time between switches or a bounded-trajectory assumption.

# 2 The mechanical energy estimate

We prove the global estimates before using any trajectory boundedness. Assumption (1) gives

$$\begin{aligned} \frac{\mu}{2}\|x-z\|^2 \le f(x) - f_* \le \frac{L}{2}\|x-z\|^2, \\ f(x) - f_* \le \frac{\|\nabla f(x)\|^2}{2\mu}, \qquad \|\nabla f(x) - \nabla f(v)\| \le L\|x-v\|. \end{aligned} \tag{16}$$

For completeness, the strong-convexity lower model at the origin makes $f$ coercive, so a minimizer exists in finite dimension and is unique. The gradient bound for the function gap follows by completing the square in the strong-convexity lower model at $x$. Thus these facts do not depend on capture of the trajectory.

**Lemma 1.** *Let $(x,p)$ be arbitrary, $\theta \in [\theta_0, 1]$, and set*

$$d = x - p, \qquad u = \theta d - s\big(\nabla f(x) - \nabla f(p)\big).$$

*Then $\|u\| \le r\|d\|$.*

**Proof.** The symmetric matrix

$$H = \int_0^1 \nabla^2 f(p + td)\, dt$$

has spectrum in $[\mu, L]$ and satisfies $\nabla f(x) - \nabla f(p) = Hd$, also when $d = 0$. Hence $u = (\theta I - sH)d$. For every eigenvalue $\nu$ of $H$,

$$\theta - s\nu \le 1 - s\mu \le r, \qquad s\nu - \theta \le sL - \theta_0 \le 1 - \theta_0 \le r.$$

The spectral theorem gives the asserted bound. Only this one averaged Hessian is diagonalized; no common eigenbasis across different phases is needed. ∎

**Lemma 2.** *The energy* (5) *satisfies* (7) *and the one-step loss* (8) *on every independent phase.*

**Proof.** Write $e = x - z$ and $d = x - p$. From (16),

$$E(x, p) \ge \frac{\gamma}{2}\|e\|^2 + \frac{1}{2}\|d\|^2.$$

Since

$$\|x - z\|^2 + \|p - z\|^2 \le 3\|e\|^2 + 2\|d\|^2,$$

the lower bound in (7) follows from $0 < \gamma \le 1$. For the upper bound, use $sL \le 1$ and

$$E(x, p) \le \frac{1}{2}\|x - z\|^2 + \frac{1}{2}\|x - p\|^2 \le \frac{3}{2}\|x - z\|^2 + \|p - z\|^2.$$

To prove the loss, write

$$y = x + u, \qquad g = \nabla f(y), \qquad x^+ = y - sg, \qquad d^+ = x^+ - x = u - sg.$$

Strong convexity at $y$ and the smooth descent inequality give

$$f(y) - f(x) \le \langle g, u\rangle - \frac{\mu}{2}\|u\|^2, \tag{17}$$

$$f(x^+) - f(y) \le -s(1 - sL/2)\|g\|^2. \tag{18}$$

Adding $s$ times these inequalities and the exact kinetic-energy change yields

$$\begin{aligned} E(x^+, x) - E(x, p) &\le s\langle g, u\rangle - \frac{\gamma}{2}\|u\|^2 - s^2(1 - sL/2)\|g\|^2 + \frac{1}{2}\|u - sg\|^2 - \frac{1}{2}\|d\|^2 \\ &= \frac{1 - \gamma}{2}\|u\|^2 - \frac{1}{2}\|d\|^2 - \frac{s^2(1 - sL)}{2}\|g\|^2 \\ &\le -a\|d\|^2 - b\|g\|^2, \end{aligned} \tag{19}$$

by Lemma 1 and $1 - \gamma \ge 0$. This proves the loss for the actual update. ∎

No estimate has divided by $\|d\|$ or $\|g\|$. Monotonicity of $E_k$ and (7) already imply

$$\|x_k - z\|^2 + \|x_{k-1} - z\|^2 \le \frac{6E_1}{\gamma} \qquad (k \ge 1). \tag{20}$$

At $sL = 1$, the coefficient $b$ vanishes. We retain the strictly positive loss in the old increment and use two successive updates.

# 3 Finite-step decrease and trajectory convergence

**Proof of Theorem 1.** Every finite-index iterate exists by explicit evaluation of the globally defined gradient. The constant bounds follow from $0 < \gamma \le 1$, $0 \le r < 1$, $0 < a \le 1/2$ and $C \ge 1/\gamma \ge 1$. Lemma 2 proves (7) and (8).

For the same arbitrary phase as in that lemma, the actual update gives $s\nabla f(y) = u - d^+$. Therefore,

$$\begin{aligned}\|s\nabla f(x)\| &\le \|s\nabla f(y)\| + sL\|x - y\| \\ &\le \|u - d^+\| + \|u\| \le 2r\|d\| + \|d^+\|.\end{aligned}$$

Using (16) and the square of this inequality,

$$\begin{aligned}E(x,p) &\le \frac{\|s\nabla f(x)\|^2}{2\gamma} + \frac{1}{2}\|d\|^2 \\ &\le \left(\frac{1}{2} + \frac{4r^2}{\gamma}\right)\|d\|^2 + \frac{1}{\gamma}\|d^+\|^2 \le C\big(\|d\|^2 + \|d^+\|^2\big).\end{aligned} \tag{21}$$

This proves (9). It bounds the present energy using two actual consecutive increments, including when either increment is zero.

Apply (8) twice, with any two admissible coefficients. Dropping only nonnegative gradient losses and using (9), we obtain

$$E_{k+2} \le E_k - a\big(\|d_k\|^2 + \|d_{k+1}\|^2\big) \le (1 - a/C)E_k = qE_k.$$

No relation between the two coefficients has entered. If $V_k = E_k + E_{k+1}$, then $E_k \le V_k \le 2E_k$ and

$$V_{k+1} - V_k = E_{k+2} - E_k \le -\rho E_k \le -\frac{\rho}{2}V_k.$$

The same argument on an arbitrary phase with independent current and next coefficients proves (11). Combining $E \le \mathcal{V}_\theta \le 2E$ with (7) proves (12). This is a uniform estimate for the family, rather than an assumption of regular switching.

Iteration of (10) on the odd and even subsequences, using $E_2 \le E_1$, gives (13). The separate nonnegative terms in $E_k$ give the first, second and fourth bounds in (14). Lemma 1 gives the remaining bound:

$$\|y_k - z\|^2 \le 2\|x_k - z\|^2 + 2\|y_k - x_k\|^2 \le 4(1/\gamma + r^2)E_k.$$

Since $q < 1$, both physical and extrapolated points converge geometrically. Finally,

$$\sum_{k=1}^{\infty}\|d_k\| \le \sqrt{2E_1}\sum_{k=1}^{\infty} q^{\lfloor (k-1)/2\rfloor/2} = 2\sqrt{2E_1}\sum_{m=0}^{\infty}(\sqrt{q})^m,$$

which proves (15). The argument applies first to finite partial sums and then to their increasing limit. If $E_1 = 0$, coercivity gives $x_0 = x_1 = z$, and the recurrence stays at that physical point. No division by initial energy is used. ∎

Two simple examples clarify the endpoint and the family notation. If $f(x) = x^2/2$, $s = 1$ and $(x, p) = (1, 1)$, then $y = 1$, $x^+ = 0$ and

$$E(x, p) = E(x^+, x) = \frac{1}{2}.$$

At the following step the new physical point is again zero, for any admissible next coefficient, and its energy is zero. Thus strict one-step decrease of this particular $E$ cannot be asserted at every nonoptimal endpoint phase. The two-step estimate remains valid.

The strict family depends on its index. For the same quadratic, now take $s = 1/2$, $\theta_0 = 1/2$ and $(x, p) = (1, 0)$. Direct evaluation gives

$$\mathcal{V}_{1/2}(1, 0) = \frac{15}{16}, \qquad \mathcal{V}_1(1, 0) = \frac{59}{64}.$$

Consequently (11) must be read as a contraction across a coefficient-indexed family. It is not a claim that (6) is the same function of $(x, p)$ for all $\theta$.

# 4 The strict speed-clock rule

We now specify one consistent original-index interpretation of the discrete rule in [1, Algorithm 1]. Fix

$$\alpha \geq 1, \qquad \beta = h > 0, \qquad h^2 L \leq 1, \qquad 0 < \lambda < \frac{1}{2\alpha}. \tag{22}$$

Use the experimental initialization $x_0 = x_1 \in \mathbb{R}^d$ and set $j_1 = 1$. At iteration $k \geq 1$, compute

$$\begin{aligned} y_k &= x_k + \frac{j_k}{j_k + \alpha}(x_k - x_{k-1}) - \beta h\big(\nabla f(x_k) - \nabla f(x_{k-1})\big), \\ x_{k+1} &= y_k - h^2 \nabla f(y_k), \end{aligned} \tag{23}$$

and only then update the clock by

$$j_{k+1} = \begin{cases} 1, & \|x_{k+1} - x_k\|^2 < \left(1 - \dfrac{2\alpha\lambda}{j_k}\right) \|x_k - x_{k-1}\|^2, \\ j_k + 1, & \text{otherwise.} \end{cases} \tag{24}$$

Both physical coordinates are retained. In particular, there is no rejected trial, recomputation at $x_k$ or assignment of two consecutive points to the same value.

**Corollary 4.1.** *Under* (1) *and* (22)*, every trajectory of* (23)–(24) *with the stated initialization satisfies all conclusions of Theorem 1, with*

$$s = h^2, \qquad \theta_0 = \frac{1}{1 + \alpha}.$$

*These conclusions include $h^2 L = 1$. No bound on the length or number of restart intervals is required.*

**Proof.** The clock is always a positive integer, since each update sends it to 1 or to $j_k + 1$. Thus

$$\frac{1}{1+\alpha} \le \theta_k := \frac{j_k}{j_k+\alpha} < 1.$$

Also $\beta h = h^2 = s$, so (23) is exactly (3). The theorem applies to its actual coefficient sequence, whatever branch occurs in (24). ∎

For the stated open interval of $\lambda$, the multiplier in the test lies strictly between zero and one. Equality always takes the no-reset branch. In particular, a zero current increment cannot trigger a reset. At the first step,

$$y_1 = x_1, \qquad x_2 = x_1 - h^2 \nabla f(x_1), \qquad j_2 = 2.$$

If $x_0 = x_1 = z$, the physical trajectory remains fixed but $j_k = k$. The energy controls the physical phase uniformly in the clock; it does not imply boundedness or convergence of the clock. Consecutive resets and infinite no-reset tails require no separate convergence argument.

The source conventions deserve precision. The box in [1, p. 14] prints $x_0 = x_{-1}$ but starts its loop at $k = 1$, which requires $x_1$. The quadratic experiment on the same page explicitly uses $x_0 = x_1$. We adopt the latter documented convention. Moreover, its display (17) uses $\le$, whereas the algorithm box uses the strict inequality retained in (24). The corollary concerns this specified interpretation, not an unreported implementation or an inference from a continuous trajectory.

# 5 Scope and source correspondence

The continuous restart theorem in [1] is distinct from the discrete energy argument here. The 2023 predecessor [5] uses the coefficient $1-\alpha/j$ and a minimum-clock safeguard; its early negative coefficients are not automatically in (2). The constant-coefficient version in [6] has a different state assignment at a speed trigger. Neither rule is silently substituted for (24).

The proved IGAHD display in [2, Section 3.2 and Theorem 6] includes an additional previous-gradient term. The discrete Wang–Peypouquet schemes [3, 4] perform a gradient step followed by extrapolation under their respective parameter relations. Bao, Chen, Li and Shen [7] analyze a gradient-angle rule which rejects and recomputes a triggering step. These precise differences prevent a direct substitution into those displayed theorems. This comparison does not exclude all possible changes of variables or all other literature. The accompanying source note records the inspected versions and their reading limits.

The content of the note is the uniform physical-phase energy estimate for (3), its endpoint finite-step decrease and the documented clock-rule corollary. The proof does not establish a maximal step range, an optimal rate, a quantitative benefit from restarting or a general convergence theorem for every Hessian-damped algorithm. The supplied

exact-arithmetic program checks finite quadratic and nonquadratic instances and several coefficient identities; the nonlinear and infinite-time conclusions follow from the proof above.

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